

§9. 一元函数积分学计算

一. 微分法

$$\int \frac{\sqrt{x}}{\sqrt{4-x^3}} dx = \int \frac{\frac{2}{3} d(x^{\frac{3}{2}})}{\sqrt{4-(x^{\frac{3}{2}})^2}} \quad x^{\frac{3}{2}}=u$$

$$= \frac{2}{3} \arcsin \frac{u}{2} = \frac{2}{3} \operatorname{arcsinh} \left(\frac{x^{\frac{3}{2}}}{2} \right) + C$$

(2020).

$$\int \frac{\operatorname{arcsinh} x}{\sqrt{x(1-x)}} dx = 2 \int \operatorname{arcsinh} x d \operatorname{arcsinh} x.$$

找谁复杂: $(\operatorname{arcsinh} x)' = \frac{1}{2\sqrt{1+x^2}} = \frac{1}{2} \frac{1}{\sqrt{x(1-x)}}$

$$I = \int \frac{\cos x - \sin x}{\cos x (1 + \cos x \cdot e^{\sin x})} dx$$

$$[\cos x \cdot e^{\sin x}]' = -\sin x \cdot e^{\sin x} + \cos x \cdot e^{\sin x} \cos x$$

$$= e^{\sin x} (\cos^2 x - \sin x)$$

$$I = \int \frac{d(\cos x \cdot e^{\sin x})}{(\cos x \cdot e^{\sin x}) (1 + \cos x \cdot e^{\sin x})} dx = \int \frac{du}{u(1+u)}$$

二. 换元法

$$\sqrt{a^2 - e^{2x}} \quad \text{令 } e^x = \frac{t}{a} \Rightarrow \sqrt{a^2 - \frac{t^2}{a^2}} = \frac{1}{a} \sqrt{a^4 - t^2}$$

• P162. 例9.7.

$$\int_0^{-\ln 2} \sqrt{1-e^{2x}} dx$$

$$\text{令 } e^x = \frac{\sin t}{\sin \frac{\pi}{6}} \quad \int_{\frac{\pi}{2}}^{\frac{\pi}{6}} \frac{\cos t}{\sin t} dt$$

$$dx = d \ln \sin t = \frac{\cos t}{\sin t} \quad x=0 = \ln \sin \frac{\pi}{6}, \quad t = \frac{\pi}{6}$$

$$x = -\ln 2 = \ln \frac{1}{2} = \ln \sin \frac{\pi}{2}, \quad t = \frac{\pi}{2}$$

$$= \int_{\frac{\pi}{2}}^{\frac{\pi}{6}} \frac{1 - \sin^2 t}{\sin t} dt = \frac{\sqrt{3}}{2} + \ln(2 - \sqrt{3})$$

三. 分部积分法. 反对幂指三.

1) $\int (x^4 + 2x + 6) e^{2x} dx.$

$$\begin{array}{ccccccc} x^4 + 2x + 6 & & 4x^3 + 2 & & 12x^2 & & 24x & & 24 & & 0 \\ e^{2x} & \times & \frac{1}{2} e^{2x} & \times & \frac{1}{4} e^{2x} & \times & \frac{1}{8} e^{2x} & \times & \frac{1}{16} e^{2x} & \times & \frac{1}{32} e^{2x} \end{array}$$

2) $\int_0^{\pi} e^{-2x} \sin x dx$ [反对幂指三]

$$\begin{array}{cccc} \sin x & & \cos x & & -\sin x \\ e^{-2x} & \times & -\frac{1}{2} e^{-2x} & \times & \frac{1}{4} e^{-2x} \end{array}$$

$$I = -\frac{1}{2} [\sin x e^{-2x}]_0^{\pi} - \frac{1}{4} [\cos x \cdot e^{-2x}]_0^{\pi} - \int_0^{\pi} \sin x e^{-2x} dx$$

eg. $\ln x$ $\arctan x$ $\arcsin x$. 求导一次就行] 4. 有理函数积分

$$\int x \ln x dx = \int \frac{\ln x}{\frac{1}{x}} dx = \frac{x^2}{2} \ln x - \int x^2 \frac{1}{2x} dx \quad \checkmark$$

例 9.27 数形结合

【例 9.27】如图 9-4 所示, 曲线 C 的方程为 $y = f(x)$, 点 $(3, 2)$ 是它的一个拐点, 直线 l_1 与 l_2 分别是曲线 C 在点 $(0, 0)$ 与 $(3, 2)$ 处的切线, 其交点为 $(2, 4)$. 设函数 $f(x)$ 具有三阶连续导数, 计算定积分 $\int_0^3 (x^2 + x) f'''(x) dx$.

【解】 $\int_0^3 (x^2 + x) f'''(x) dx$
 $= (x^2 + x) f''(x) \Big|_0^3 - \int_0^3 (2x + 1) f''(x) dx$

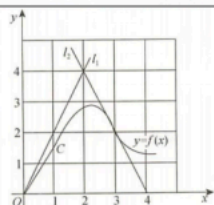


图 9-4

$$\int_0^3 (x^2 + x) f'''(x) dx$$

$$\begin{cases} f(0)=0 & f''(0)=f'(3)=2 \\ f'(0)=2 & f'(3)=-2 \\ f''(3)=0 \end{cases}$$

$$\int_0^3 (x^2 + x) f'''(x) dx$$

$$\begin{aligned} & \begin{matrix} x^2+x & 2x+1 & 2 & 0 \\ f'''(x) & f''(x) & f'(x) & f(x) \end{matrix} \\ & \left[(x^2+x) f''(x) - (2x+1) f'(x) + 2f(x) \right] \Big|_{x=0}^{x=3} \\ & = 0 - (7 \cdot (-2) - 2) + 2(2-0) = 20 \end{aligned}$$

例 1. $\int \frac{x^4+1}{x(x^2+1)^2} dx$

$$\text{令 } x^2=t \quad \frac{dx}{dt} = \frac{1}{2\sqrt{t}} \Rightarrow \frac{1}{2} \int \frac{t^2+1}{t(t+1)^2} dt$$

$$= \frac{1}{2} \left(\frac{A}{t} + \frac{B}{t+1} + \frac{C}{(t+1)^2} \right) \quad \left. \begin{aligned} \text{令 } t=0, & \quad A=1 \\ A+B=1 & \quad B=0 \\ \text{令 } t=-1, & \quad -C=2 \quad C=-2 \end{aligned} \right\}$$

习 9.11

$$\int \frac{1}{e^{2x} + 3e^x + 2} dx \quad \begin{matrix} \text{令 } x=t \\ dt=e^x dx \end{matrix} \Rightarrow \int \frac{dt}{t^2+2t+2}$$

5. 区间再现公式.

$$\left[\begin{aligned} \text{证明 } \int_a^b f(x) dx &= \int_a^b f(a+b-x) dx. \\ \text{换元法. 令 } x &= a+b-t \\ \int_a^b f(x) dx & \xrightarrow{x=a+b-t} \int_b^a f(a+b-t) (-dt) \end{aligned} \right]$$

例 9.9

$$\begin{aligned} I &= \int_0^1 \frac{\ln(1+x)}{1+x^2} dx \quad \begin{matrix} 1+x^2 \\ x = \tan t \end{matrix} \Rightarrow \int_0^{\frac{\pi}{4}} \frac{\ln(1+\tan t)}{\sec^2 t} dt \cdot \tan t \\ &= \int_0^{\frac{\pi}{4}} \frac{\ln(1+\tan(0+\frac{\pi}{4}-t))}{\sec^2 t} dt = \int_0^{\frac{\pi}{4}} \ln \left[1 + \frac{\tan \frac{\pi}{4} - \tan t}{1 + \tan \frac{\pi}{4} \tan t} \right] dt \end{aligned}$$

$$= \int_0^{\frac{\pi}{2}} \ln \frac{2}{1+\tan t} dt = \int_0^{\frac{\pi}{2}} \ln 2 dt - \int_0^{\frac{\pi}{2}} (1+\tan t) dt$$

$$I = \frac{\pi}{8} \ln 2$$

6. 点火公式

基本公式 $\int_0^{\frac{\pi}{2}} \rightarrow \frac{1}{2} \cdot \frac{\pi}{2} / \frac{2}{2}$

扩展公式 $\int_0^{\pi}$ double. $\int_0^{\pi} \sin x \leftarrow \int_0^{\pi} \cos x \leftarrow \begin{matrix} \text{奇} 0 \\ \times 2 \end{matrix}$

$\int_0^{2\pi}$ double x double $\int_0^{2\pi} \sin x = \int_0^{2\pi} \cos x \leftarrow \begin{matrix} \text{奇} 0 \\ \times 4 \end{matrix}$

例 9.16

求 $I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^4 x}{1+e^{-x}} dx$.

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^4 x}{1+e^{-x}} dx = \int_{-\frac{\pi}{2}}^0 \frac{\sin^4 x}{1+e^{-x}} dx + \int_0^{\frac{\pi}{2}} \frac{\sin^4(-x)}{1+e^x} dx$$

$\xrightarrow{a+b \rightarrow a, b}$ $\xrightarrow{f(x) \rightarrow f(a+b-x)}$

$$= \int_{-\frac{\pi}{2}}^0 \frac{(e^x+1)\sin^4 x}{1+e^x} dx = \int_{-\frac{\pi}{2}}^0 \sin^4 x dx \xrightarrow{x=-t} \int_0^{\frac{\pi}{2}} \sin^4 t dt$$

$$= \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

例 9.17

证 $\int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx$

$$\int_0^{\pi} x f(\sin x) dx = \frac{1}{2} \int_0^{\pi} x f(\sin x) + (0+\pi-x) f(\sin(0+\pi-x)) dx$$

$$= \frac{1}{2} \int_0^{\pi} x f(\sin x) + \pi f(\sin x) - x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx$$

例 9.18

证 $\int_0^{\pi} x f(\sin x) dx = \pi \int_0^{\frac{\pi}{2}} f(\sin x) dx$.

$$\int_0^{\pi} x f(\sin x) dx = \int_0^{\frac{\pi}{2}} (0+\pi-x) f(\sin(0+\pi-x)) + x f(\sin x) dx$$

$$= \pi \int_0^{\frac{\pi}{2}} \sin x f(\sin x) dx \quad \overline{f(x)} \perp$$

如 $\int_0^{\pi} x \sin^9 x dx = \pi \int_0^{\frac{\pi}{2}} \sin^9 x dx$

$$= \frac{8}{9} \cdot \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} \cdot 1$$

例 9.21

求 $I = \int_0^{\frac{\pi}{2}} \frac{\sin^3 x}{\sin x + \cos x} dx$

$$= \frac{1}{2} \left[\int_0^{\frac{\pi}{2}} \frac{\sin^3 x}{\sin x + \cos x} dx + \int_0^{\frac{\pi}{2}} \frac{\sin^3 (0+\frac{\pi}{2}-x)}{\sin(0+\frac{\pi}{2}-x) + \cos(0+\frac{\pi}{2}-x)} dx \right]$$

$$= \frac{1}{2} \left[\int_0^{\frac{\pi}{2}} \frac{\sin^3 x}{\sin x + \cos x} + \frac{-\cos^3 x}{-\cos x - \sin x} dx \right]$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{\sin^3 x + \cos^3 x}{\sin x + \cos x} dx = \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin^2 x - \sin x \cos x + \cos^2 x dx$$

7. 区间能简化公式

$$\int_a^b f(x) dx \quad x = \frac{a+b}{2} + \frac{b-a}{2} \sin t \quad \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} f\left(\frac{a+b}{2} + \frac{b-a}{2} \sin t\right) \cdot \frac{b-a}{2} \cos t dt$$

例 9.22

$$I = \int_1^3 \frac{dx}{\sqrt{(3-x)(x-1)}} \quad \text{令 } x = \frac{1+3}{2} + \frac{3-1}{2} \sin t = (\sin t) + 2$$

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos t dt}{\sqrt{(3-2-\sin t)(\sin t+2-1)}} = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sqrt{1-\sin^2 t} dt}{\sqrt{1-\sin^2 t}} = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1 dt = \pi$$

$$\text{推广: } \int_a^b \frac{dx}{\sqrt{(x-a)(b-x)}} = \pi$$

例 9.23

$$I = \int_1^3 \sqrt{(3-x)(x-1)} dx \quad \text{令 } x = 1 + (3-1)t = 2t+1$$

$$I = \int_0^1 \sqrt{(2-2t)(2t)} \cdot 2 dt = \int_0^1 4 \sqrt{t(1-t)} dt$$

$$= 4 \int_0^1 \sqrt{t-t^2} dt = 4 \int_0^1 \sqrt{\left(\frac{1}{2}\right)^2 - \left(t-\frac{1}{2}\right)^2} dt$$

$$= 4 \cdot \frac{\pi}{8} = \frac{\pi}{2} \quad \text{根据: } \int \sqrt{a^2-x^2} dx = \frac{a^2}{2} \arcsin \frac{x}{a} + \frac{x}{2} \sqrt{a^2-x^2}$$

$$\text{例 2} \quad \text{令 } x = \frac{1+3}{2} + \frac{3-1}{2} \sin t = \sin t + 2$$

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{(1-\sin^2 t)(\sin t+1)} \cos t dt \\ = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{1-\sin^2 t} \cos t dt = 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 t dt \\ = 2 \int_0^{\frac{\pi}{2}} \cos^2 t dt = 2 \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{2}$$

8. 对称性下的积分问题

eg 例 9.25(1)

$$\int_0^4 x \sqrt{4x-x^2} dx = \int_0^4 x \sqrt{x(x-4)} dx \quad (x-2)^2 = x^2 - 4x + 4 \\ - (x-2)^2 = -x^2 + 4x - 4 \\ x-2=t \quad \int_{-2}^2 (t+2) \sqrt{2^2-t^2} dt = \int_{-2}^2 t \sqrt{2^2-t^2} dt + \int_{-2}^2 2 \sqrt{2^2-t^2} dt \\ = 2 \int_{-2}^2 \sqrt{2^2-t^2} dt = 4\pi$$


eg 例 9.25(2)

$$\int_0^2 (2x+1) \sqrt{2x-x^2} dx \quad x = x+1-1 \quad t = x-1 \\ = \int_0^2 (2x+2-1) \sqrt{1-(x-1)^2} dx = \int_{-1}^1 (2x+1) \sqrt{1-t^2} dt \\ = \int_{-1}^1 (2(x+1)-1) \sqrt{1-t^2} dt = \int_{-1}^1 (2x+1) \sqrt{1-t^2} dt = \frac{\pi}{2}$$

七. 附例 (积分) 9.26 附例 (求导) 9.28

例 9.26

$g(x)$ 一个偶函数 $\ln(x+1)$

$$f(x) = \lim_{t \rightarrow \infty} t^2 \sin \frac{x}{t} \cdot [g(2x + \frac{1}{t}) - g(2x)] \quad \int_0^1 f(x) dx?$$

[solution]

$$f(x) = \lim_{t \rightarrow \infty} \frac{\sin \frac{x}{t}}{\frac{x}{t}} \cdot x \cdot \frac{g(2x + \frac{1}{t}) - g(2x)}{\frac{1}{t}}$$

$$\lim_{t \rightarrow \infty} \frac{\sin \frac{x}{t}}{\frac{x}{t}} \Rightarrow 1. \quad \lim_{t \rightarrow \infty} \frac{g(2x + \frac{1}{t}) - g(2x)}{\frac{1}{t}} = g'(2x)$$

$$\therefore f(x) = x g'(2x)$$

$$g(x) = (\ln(x+1))' = \frac{1}{x+1} \quad g'(2x) = (\frac{1}{2x+1})' = -\frac{1}{(2x+1)^2}$$

$$\int_0^1 f(x) dx = \int_0^1 \frac{x}{(2x+1)^2} dx = \int_0^1 \frac{x}{2} d \frac{1}{2x+1} = \frac{1}{2} \left[\frac{x}{2x+1} \right]_0^1 - \int_0^1 \frac{1}{(2x+1)^2} dx$$

$$= \frac{1}{2} \left[\frac{1}{3} - \frac{1}{2x+1} \right]_0^1 = \frac{1}{6} - \frac{\ln 2}{4}$$

例 9.38

(换序型)

$$\text{设 } f(x) = \int_0^x \frac{\sin t}{\pi - t} dt. \quad \text{求 } \int_0^\pi f(x) dx$$

$$\Rightarrow f'(x) = -\frac{\sin x}{\pi - x}, \quad x \in (0, \pi).$$

$$\int_0^\pi f(x) dx = x f(x) \Big|_0^\pi - \int_0^\pi x df(x) = \pi f(\pi) - \int_0^\pi x \cdot \frac{\sin x}{\pi - x} dx$$

$$\Rightarrow \int_0^\pi f(x) dx = \pi \int_0^\pi \frac{\sin x}{\pi - x} dx - \int_0^\pi x \frac{\sin x}{\pi - x} dx$$

$$= \int_0^\pi (\pi - x) \frac{\sin x}{\pi - x} dx = 2$$

+ 分段函数的定积分. [指定区间]

例 9.29.

$$\text{设 } f(x) = \begin{cases} e^{-x} & x \geq 0 \\ 1+x^2 & x < 0 \end{cases} \quad \text{求 } \int_{-2}^2 f(x-1) dx$$

$$\int_{-2}^2 f(x-1) dx \xrightarrow{x-1=t} \int_{-3}^1 f(t) dt = \int_{-3}^0 (1+t^2) dt + \int_0^1 e^{-t} dt$$

$$= \int_{-3}^0 (1+t^2) dt + \int_0^1 e^{-t} dt = 13 - \frac{1}{e}$$

十一. 分段函数变限积分. 例 30.31

例 9.39

$$f(x) = \begin{cases} 2x + \frac{3}{2}x^2 & -1 \leq x \leq 0 \\ \frac{x e^x}{(e^x + 1)^2} & 0 \leq x \leq 1 \end{cases} \quad \text{求 } F(x) = \int_{-1}^x f(t) dt \text{ 表达式}$$

① $x \in (-1, 0)$

$$F(x) = \int_{-1}^x f(t) dt = \int_{-1}^x (2t + \frac{3}{2}t^2) dt$$

② $x \in [0, 1]$

$$f(x) = \int_{-1}^0 (2t + \frac{3}{2}t^2) dt + \int_0^x \frac{t e^t}{(e^t + 1)^2} dt.$$

十二. 四大方法

12.1 直接求导型 → 用“巧劲”

例 9.33

设 $f(x)$ 在 $[0, +\infty)$ 上可导, $f(0)=0$.

且其反函数为 $g(x)$. 若 $\int_0^{f(x)} g(t) dt = x^2 e^x$ 求 $f(x)$

$$\left[\int_0^{f(x)} g(t) dt \right]' = \underbrace{g[f(x)]}_{x} \cdot f'(x) - 0$$

$$= x f'(x)$$

$$[x e^x]' = 2x e^x + x^2 e^x$$

$$x \neq 0 \text{ 时 } f'(x) = e^x (2+x) \quad f(x) = \int e^x (2+x) dx$$

$$\begin{matrix} 2+x & 1 & 0 \\ e^x & e^x & e^x \end{matrix} = (2+x)e^x - e^x + C$$

$$x=0 \text{ 时 } 0 = f(0) = \lim_{x \rightarrow 0^+} f(x) = 1 + C \Rightarrow C = -1$$

$$\Rightarrow f(x) = (2+x)e^x - e^x - 1$$

12.2 换元求导型

例 9.34

$f(x)$ 在 $[0, +\infty)$ 可导, $f(0)=0$. 反函数 $g(x)$.

$$\int_x^{x+f(x)} g(t-x) dt = x^2 \ln(1+x) \quad \text{求 } f(x).$$

Solution:

$$\int_x^{x+f(x)} g(t-x) dt = x^2 \ln(1+x) \quad \boxed{x \text{ 是常数}}$$

$$\int_x^{x+f(x)} g(t-x) dt = \int_{t=x}^{t=x+f(x)} g(t-x) dt$$

$$\begin{matrix} t=x: u=t-x=0 \\ t=x+f(x): u=f(x) \end{matrix}$$

$$\int_0^{f(x)} g(u) du = x^2 \ln(1+x)$$

↓ 求导 ↓

$$x f'(x) = f'(x) \cdot g[f(x)] = 2x \ln(1+x) + \frac{x^2}{1+x}$$

$$x \neq 0 \quad f'(x) = 2 \ln(1+x) + \frac{x}{1+x} \quad f(x) = \ln(1+x) + 2x \ln(1+x) - x + C$$

$$x=0 \text{ 时 } \lim_{x \rightarrow 0^+} f(x) = f(0) = 0 \Rightarrow C = 0$$

$$\Rightarrow f(x) = \ln(1+x) + 2x \ln(1+x) - x$$

例 9.35

$f(x)$ 在 $(-\infty, +\infty)$ 非恒连续, 且

$$f(x) \int_0^x f(x-t) dt = \sin^4 x$$

求 $f(x)$ 在 $[0, \pi]$ 上平均值

(solution)

$$\bar{f}(x) = \frac{\int_0^x f(x) dx}{x - 0} = \frac{\int_0^x f(x-t) dt}{x-t=u} \int_x^0 f(u) - du = \int_0^x f(u) du$$

$$\Rightarrow f(x) \cdot \int_0^x f(u) du = \sin^4 x$$

$$\int_0^\pi f(x) dF(x) = \int_0^\pi \sin^4 x = \frac{1}{2} F^2(x) + C \Big|_0^\pi$$

$$\therefore \frac{1}{2} F^2(\pi) = \frac{3}{8} \pi \quad F^2(\pi) = \frac{3}{4} \pi$$

$$\bar{f}(x) = \frac{[F(\pi)]}{\pi - 0} = \frac{\frac{3}{4} \pi}{\pi} \checkmark$$

(2.3. 拆分求导型.

例 9.36.

设 $|t| \leq 1$ 求 $I(t) = \int_{-1}^1 |x-t| e^{2x} dx$ 最大值

[当 x, t 同-区间时. $-1 \xrightarrow{t} 1$, 用 t 划为区间]

$$I(t) = \int_{-1}^1 = \int_{-1}^t (t-x) e^{2x} dx + \int_t^1 (x-t) e^{2x} dx$$

例 9.37

设 $f(x) = \int_0^1 |t^2 - x^2| dt \quad (x > 0)$ 求 $f'(x)$, 并求 $f(x)$ min.

[当 x, t 在不同区间, $0 \xrightarrow{t} 1$, 分情况讨论]

当 $x \in (0, 1)$ $\rightarrow x$ 与 t 变成同-区间.

$$f(x) = \int_0^x (x^2 - t^2) dt + \int_x^1 (t^2 - x^2) dt$$

当 $x \in (1, +\infty)$

$$f(x) = \int_0^1 (x^2 - t^2) dt$$

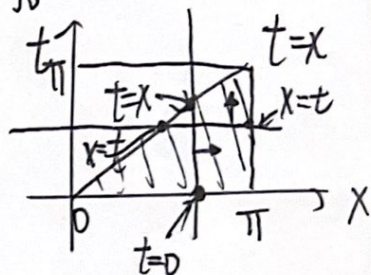
(2.4 换序型.

设 $f(x) = \int_0^x \frac{\sin t}{\pi - t} dt$, 求 $\int_0^\pi f(x) dx$.

$$\Rightarrow \int_0^\pi \left(\int_0^x \frac{\sin t}{\pi - t} dt \right) dx$$

$$= \int_0^\pi \left(\int_x^\pi \frac{\sin t}{\pi - t} \cdot dx \right) dt$$

$$= \int_0^\pi \frac{\sin t}{\pi - t} \cdot (\pi - t) dt = \int_0^\pi \sin t dt = 2$$



十三. 反常积分的计算.

收敛时: 定积分 + 极点.

例 9.39

$$\int_5^{+\infty} \frac{1}{x^2 - 4x + 3} dx = ?$$

$$= \int_5^{+\infty} \frac{1}{(x-1)(x-3)} dx = \frac{1}{2} \ln \frac{x-3}{x-1} \Big|_5^{+\infty} = \frac{1}{2} \ln 2$$

例 9.40 $\leftarrow$ 换元法

$$\int_0^1 \frac{x^2 \operatorname{arcsinh} x}{1-x^2} dx = \int_0^1 x^2 \operatorname{arcsinh} x d \operatorname{arcsinh} x$$

$$\stackrel{t = \operatorname{arcsinh} x}{=} \int_0^{\frac{\pi}{2}} x \sin x \cdot dx = \int_0^{\frac{\pi}{2}} \sin x dx$$

$$= \frac{x + \frac{1}{2} x \sin 2x}{2} \Big|_0^{\frac{\pi}{2}} - \frac{\frac{x}{2} - \frac{1}{4} \cos 2x}{2} \Big|_0^{\frac{\pi}{2}} = \frac{\pi}{16} + \frac{1}{4}$$